FACT Course Breakdown

Before deciding whether AI should be allowed, identify what kind of learning is being assessed.

C - Conceptual Understanding

Typical AI role: No AI or AI-resistant

This is what students must understand deeply enough to explain, transfer, and apply across contexts.

C is what students must understand, not just produce.

Examples:

1. Explain why an equation works
2. Explain causation in history
3. Explain how evidence supports a claim
4. Explain a biological process
5. Explain why a method works
6. Explain when a method is appropriate
7. Interpret a theory
8. Explain assumptions behind a model

F - Fundamental Skills

Typical AI role: No AI / AI-resistant

This is what students must be able to do without AI assistance because it forms the foundation for later learning.

F is the skill students need before AI becomes useful rather than harmful.

Examples:

1. Solve a basic equation
2. Read a primary source
3. Write a paragraph with evidence
4. Identify parts of a cell
5. Perform a basic lab procedure
6. Use a core disciplinary method
7. Explain a foundational term
8. Create a basic diagram or prototype

T - critical Thinking

Typical AI role: Structured AI use

This is where students must make judgments: evaluate evidence, critique AI output, defend decisions, interpret uncertainty, or synthesize ideas.

T is where students must show judgment.

Examples:

1. Evaluate whether an AI-generated answer is accurate
2. Identify bias in a historical interpretation
3. Revise AI output using disciplinary knowledge
4. Explain uncertainty in a biological explanation
5. Decide what evidence is trustworthy
6. Critique a model prediction
7. Compare competing interpretations
8. Defend a design decision

A - Applied Work

Typical AI role: Structured or open AI use

This is where students use knowledge in a real-world, authentic, or professional context.

A is where AI can help students do more authentic work, as long as the student remains responsible for the outcome.

Examples:

1. Analyze a real dataset
2. Translate or summarize archival material
3. Draft a policy memo
4. Interpret clinical or case-study information
5. Analyze lab or field data
6. Build a model
7. Apply the theory to a real case
8. Develop a prototype or design

The numbers align across cells. For example, 1 quantitative reasoning, 2 historical sources, 3 evidence-based writing, 4 biological systems, 5 lab or field methods, 6 disciplinary methods and modeling, 7 theory application, and 8 design, modeling, or prototyping

FACT framework adapted from Elshall and Badir (2025)

Activity 1: Break One Topic into FACT

Faculty first map learning before deciding AI use.

Blank Faculty Worksheet

Choose one topic you teach, then complete the table.

My topic: ____

FACT component	Example	Assessment Weight	AI role
C - Concept *What must students understand?*			
F - Foundation *What must students do independently?*			
T - Thinking *What must students judge or critique?*			
A - Application *Where could students apply this topic?*			

AI roles to consider: No AI | AI-resistant | Structured AI | Open AI

Final reflection: Where should AI be limited, structured, or opened - and why?

Final Note: This activity supports assignment design. Implementation still requires clear instructions, checkpoints, rubrics, and verification strategies.

Solved Example 1: Groundwater Hydrology

Topic: Darcy's Law and groundwater flow

FACT component	Example	Assessment Weight	AI role
C - Concept *What must students understand?*	Groundwater flows from high hydraulic head to low hydraulic head.	50%	No AI
F - Foundation *What must students do independently?*	Calculate hydraulic gradient and groundwater flux using Darcy's Law.	25%	No AI
T – Thinking *What must students judge or critique?*	For Nubian sandstone aquifer, estimate age of groundwater from recharge to well at the Kharga Oases based on the provided hydraulic head contour data. For the hydraulic conductivity and porosity value provide a reliable source. Clearly explain all your assumptions and solution steps.	20%	Structured AI
A - Application *Where could students apply this topic?*	List one application of Darcy's law that is directly related to your work or study. What do you still need to learn to be able to solve this problem?	5%	Open AI

Final reflection:

- This topic is mainly a C/F-heavy topic because students must first understand hydraulic head and apply Darcy's Law independently. A broader groundwater-modeling task would place more weight on T and A.
- The topic stays the same, but the learning target changes; therefore, AI use should change with the learning target.

Solved Example 2: Literature Search for a Thesis Topic

Topic: Finding relevant literature related to my research question

Task: Students find 10 relevant sources using Web of Science and 2 additional sources using AI.

FACT component	Example	Assessment Weight	AI role
C - Concept *What must students understand?*	Understand relevance, credibility, keywords, and inclusion/exclusion logic.	25%	Structured AI
F - Foundation *What must students do independently?*	Use Web of Science to search, filter, and select relevant peer-reviewed sources.	25%	AI-resistant
T –Thinking *What must students judge or critique?*	Judge whether sources are truly relevant and whether AI-suggested sources are accurate, useful, or redundant.	25%	Structured AI
A - Application *Where could students apply this topic?*	Use Web of Science to identify 10 sources. Use AI to identify 2 additional sources. Comment on difference between two approaches	25%	Structured AI with verification

Final reflection:

- In higher-order activities such as literature search, the boundaries among C, F, T, and A are often blurred.
- Students should learn how to search independently first, then use AI to extend, refine, and organize the literature search - not to replace scholarly judgment.

Final Note: This activity supports assignment design. Implementation still requires clear instructions, checkpoints, rubrics, and verification strategies.

Assessment Checkpoints

A faculty menu for making student thinking visible

After identifying learning goals (F, A, C, T), choose how AI should or should not be used. ***Modality key:*** *IP = In-person | Sync = Online synchronous | Async+ = Online asynchronous with verification, process evidence, or follow-up | All = IP, Sync, or Async+*

No AI Use

Use when students must show independent understanding, recall, reasoning, or skill.

Instructor menu:

1. In-class paper quiz or exam [IP]
2. Live conceptual questions that ask students to explain why [IP, Sync]
3. Oral explanation, mini-viva, or live follow-up question [IP, Sync]
4. Whiteboard, paper, or shared-screen problem solving [IP, Sync]
5. Timed quiz with randomized conceptual questions [IP, Sync]
6. Live interpretation of a graph, source, equation, case, or concept [IP, Sync]
7. Closed-resource lab, coding, or methods demonstration [IP, Sync]
8. Asynchronous no-AI task paired with a short verification question [Async+]

Bottom line: *No AI use works best when AI access is limited or when follow-up verification checks student ownership.*

AI-Resistant Assignment

Use when students need to show process, context, and skill development - not only a final answer.

Instructor menu:

1. Personalize the task with student-specific topics, datasets, cases, sites, or constraints [All]
2. Require local, field, lab, workplace, community, or course-specific evidence [All]
3. Ask for process evidence: notes, search terms, sketches, calculations, source matrix, code history, or failed attempts [All]
4. Use staged submissions: topic > evidence > draft > revision > final [All]
5. Require version history, tracked changes, or a short work-progress log [All]
6. Ask students to explain what they did, why they did it, and what alternatives they considered [All]
7. Require a revision memo: what changed, why, and based on what feedback or evidence [All]
8. Add a short follow-up question, oral check, or screen-recorded walkthrough [IP, Sync, Async+]

Bottom line: *AI-resistant does not mean AI-proof. It means the assignment is designed so students must show context, process, verification, and judgment.*

Structured AI Use

Use when AI is allowed, but students must evaluate, critique, revise, verify, or justify its output.

Instructor menu:

1. Ask AI to generate an answer, then have students critique and correct it [All]
2. Have students compare two AI responses and justify which is stronger [All]
3. Require students to verify AI-generated claims with credible sources [All]
4. Use a prompt > output > critique > revision sequence [All]
5. Ask students to identify what AI missed, oversimplified, or got wrong [All]
6. Ask students to explain where AI might mislead a novice [All]
7. Let students use AI for brainstorming, then require them to defend their final decision [All]
8. Require an AI-use appendix showing prompts, outputs, edits, and rejected suggestions [All]

Bottom line: *Structured AI use should reward judgment, verification, revision, and explanation - not simply the production of AI output.*

Open AI Use

Use when students are completing authentic, complex, real-world, or professional work.

Instructor menu:

1. Allow full AI-supported projects with documentation [All]
2. Use AI-assisted coding, modeling, analysis, visualization, writing, or design [All]
3. Require students to disclose how AI was used and what they changed [All]
4. Require a project log, design journal, or decision memo [All]
5. Ask students to explain key choices, tradeoffs, assumptions, and limitations [All]
6. Assess process, evidence, interpretation, and final product [All]
7. Use presentations, recorded walkthroughs, peer review, or Q&A to verify ownership [IP, Sync, Async+]
8. Require students to defend conclusions and explain why the final work is trustworthy [All]

Bottom line: *Open AI use works best when students remain accountable for decisions, evidence, interpretation, and the quality of the final product.*

Adapted from Elshall and Badir (2025)

Asynchronous Safeguards

Designing visible evidence of process, judgment, and ownership

For online asynchronous courses, do not rely on "no AI" unless there is some verification. Stronger designs make student process, judgment, and ownership visible while reducing the value of generic AI output.

1. Make generic AI output less useful

These safeguards reduce the value of copy-and-paste AI responses by requiring students to connect the work to a specific context, dataset, experience, or prompt.

Safeguard	Brief explanation	Why it helps
Personalization	Ask students to connect the task to their own topic, project, experience, case, dataset, or local context.	Generic AI output becomes less useful because the response must fit the student's specific situation.
Randomized prompts	Give students slightly different questions, datasets, cases, roles, or constraints.	Reduces copying, recycled answers, and reusable AI-generated responses.
Local/context-specific data	Require students to use course-specific, local, recent, or instructor-provided information.	Students must reason with situated evidence that may not be easily generated by AI alone.

2. Make the learning process visible

These safeguards require students to show how their work developed, not just submit a final product.

Safeguard	Brief explanation	Why it helps
Process evidence	Ask students to submit outlines, notes, calculations, annotations, search terms, planning documents, or intermediate steps.	Students must show how the work was developed.
Version history	Use platforms that preserve revision history, such as Google Docs, Word Online, GitHub, or LMS drafts.	Sudden, unexplained final products become more visible.
Draft-revision trail	Require at least one draft, feedback step, and revised version.	Shows development over time and makes learning progress visible.

3. Verify evidence and accuracy

These safeguards shift the task from producing answers to checking whether claims, sources, and outputs are credible.

Safeguard	Brief explanation	Why it helps
Source verification	Require students to verify claims using real sources, citations, datasets, or course materials.	AI claims must be checked against evidence rather than accepted uncritically.
Decision memo	Ask students to explain key choices, assumptions, rejected alternatives, and final decisions.	Assesses judgment, not just the final output.
Evidence annotation	Ask students to mark where evidence supports each major claim, calculation, or design choice.	Makes reasoning and source use easier to evaluate.

4. Check ownership without full proctoring

These safeguards give instructors a lightweight way to confirm that students understand and can explain their own work.

Safeguard	Brief explanation	Why it helps
Brief screen recording	Ask students to record a short explanation of their work, process, or final decision.	Students must explain their own work in their own words.
Follow-up question	Ask one short individualized question after submission, based on the student's work.	Checks ownership without requiring full proctoring.
Reflection statement	Ask students to explain what they learned, what was difficult, and how they used or did not use AI.	Encourages metacognition and helps identify whether students understand the work.

How to use the safeguards

Faculty do not need to use every safeguard. A strong asynchronous assignment may combine two or three safeguards, such as:

- Personalized prompt + process evidence + follow-up question
- Source verification + decision memo + draft-revision trail

Take-home: In asynchronous courses, design visible evidence of process, judgment, and ownership.

Activity 2: Redesign An Assignment Checkpoint

From FACT map to assignment instruction

Purpose:
Use your Activity 1 FACT map to revise one assignment checkpoint so students must show evidence of learning.
Activity 1 identified the topic, FACT component, assessment weight, and AI role.
Activity 2 turns that design decision into clear assignment language.

Blank Faculty Worksheet

Topic from Activity 1:

Step	Your Answer
FACT row selected	C / F / T / A
AI role selected	No AI / AI-resistant / Structured AI / Open AI
What students must show	Process / Judgment / Accountability
Checkpoint added	
Revised assignment instruction	

Final reflection: What student thinking is now more visible?

Example 1: Groundwater Hydrology

Topic from Activity 1: Darcy's Law and groundwater flow

FACT row selected: T - Thinking

AI role selected: Structured AI

Step	Solved example
What students must show	Judgment about assumptions, uncertainty, and source reliability
Checkpoint added	• Assumption check • source verification • explanation of solution steps
Revised assignment instruction	• For the Nubian sandstone aquifer, estimate groundwater age from recharge to the Kharga Oases well using the provided hydraulic-head contour data. • You may use AI to help organize your calculation or check your reasoning, but you must identify reliable sources for hydraulic conductivity and porosity, explain your assumptions, show your solution steps, and state what makes the estimate uncertain.
Why this is strong AI-era assignment: AI can support the work, but students must still identify valid source, justify values, explain assumptions, and evaluate uncertainty.	

Example 2: Literature Search for a Research Question

Topic from Activity 1: Finding relevant literature related to my research question

FACT row selected: A/T - Application and Thinking

AI role selected: Structured AI with verification

Step	Solved example
What students must show	Search process, relevance judgment, and verification of AI-suggested sources
Checkpoint added	• Source matrix • AI-source verification memo
Revised assignment instruction	• Find 10 relevant peer-reviewed sources using Web of Science. • Then use AI to suggest 2 additional sources. • For all 12 sources, complete a source matrix explaining how you found each source, why it is relevant, its key idea, and one strength or limitation. • For the 2 AI-suggested sources, verify that they exist and explain whether they are useful, misleading, or redundant.

Source matrix example

Source	How I found it	Why it is relevant	Key idea	Strength / limitation
Author, year, title	Web of Science / AI / citation chasing	Connects to my research question because…	Main finding or argument	Strong method, but limited context…
Why this is strong AI-era assignment: AI is allowed, but students must demonstrate independent database searching, source evaluation, verification of AI-suggested sources, and scholarly judgment.				

Final note: A strong AI-era assignment does not just state an AI policy. It designs evidence of learning: what students did, why they did it, how they verified it, and what judgment they used.

7. Societal implications: Higher education in the AI era

7.1 Disruptive innovation

While knowledge creation was once dominated by humans, we are now increasingly seeing a shift towards human-AI co-creation (Jain et al., 2025; Lee et al., 2024; Lim et al., 2023; O'Dea, 2024; Wu & Chen, 2025). Consequently, the integration of AI in higher education may require a shift from traditional content delivery to managing co-agency between humans and machines (Gonsalves, 2024; Jain et al., 2025; Molenaar, 2022; Philbin, 2023). Other opportunities include AI as co-teacher (Niloy et al., 2025), human-AI co-creation of knowledge (Yuwono et al., 2024), and educator-learner co-design of AI-mediated learning (Carvalho et al., 2022). This may be contributing to a new paradigm of human-AI collaboration where pedagogy may need to evolve, for example, to teach students how to question, verify, and ethically manage AI-generated content (Fu & Weng, 2024; Gonsalves, 2024; Nguyen, 2025). With each major technological advance, students may need to develop new competencies not only in technical proficiency, but also in higher-order skills such as critical thinking, delegation, evaluation, and epistemic responsibility (Gonsalves, 2024; Melisa et al., 2025; Xia et al., 2024). Table 4 illustrates this pedagogical evolution, showing how pedagogical design can adapt across increasing levels of human-AI collaboration.

7.2 Designing for human expertise

While current generation models (i.e., multimodal AI and early agents) fail as the complexity of the task increases (Shojaee et al., 2025) and the precise future of AI affordances is unclear, the hypothetical example of a future autonomous scientific agent in Table 4 serves as a conceptual illustration rather than an empirically tested outcome. As scientific agents continue to evolve (Ren et al., 2025), this endpoint in Table 4 or comparable examples (Tang et al., 2025) may help inform long-term pedagogical strategies including development of meta-competencies and re-evaluation of human expertise in the AI era. These meta-competences include ethical grounding and goal alignment to ensure that as humans delegate tasks to AI, they maintain strategic and responsible oversight (Chaffer et al., 2025). This re-evaluation of human expertise includes shifting focus from routine tasks, which can be automated, to "durable competencies" that are uniquely human, such as creative problem-solving, strategic thinking, and ethical judgment (AI Alliance, 2024). In addition, the re-evaluation of human expertise is not merely an economic question but may also be interpreted as an existential one (Edelman, 2025; Kulveit et al., 2025). This opens a potential research domain focused on redefining the nature of uniquely human expertise and ethical grounding in a digital world with advanced automation (Elshall et al., 2022).

Table 4. Conceptual illustration of AI capability progression and associated pedagogical shifts

AI Affordances	Description	Example	Pedagogical Design
Pre-generative AI (Baseline)	Tools for calculation or information retrieval with human performing all synthesis and analysis	Using Google Search to find EPA datasets or using statistical software to run a pre-programmed regression analysis	Core competencies: Foundational information literacy, manual research skills, data interpretation, and proficiency with specific software
Foundational AI (c. 2022-2024)	AI as a conversational tool for brainstorming, summarizing text, and generating codes with human directing each step	Asking AI to write a Python code to plot pre-cleaned data or perform regression analysis	Builds on core competencies: Effective prompt engineering, critical evaluation of AI outputs, and understanding AI ethics
Multimodal AI and early agents (c. 2024-Present)	AI as an interactive workbench capable of analyzing data, interpreting images, and executing multi-step workflows with a single command	Uploading a raw CSV of water quality data and asking the AI to clean, analyze, and visualize the trends of nitrate levels over time	Expands to include*: Designing and validating multi-step analytical workflows, interpreting complex AI-generated analyses, and debugging complex AI processes
Autonomous scientific agents (hypothetical future)	AI acts as a research collaborator, capable of independently designing and executing complex projects, from hypothesis to conclusion	Providing a high-level directive: Investigate the impact of recent urbanization on local watershed and propose mitigation strategies	Additionally: Meta-competencies and even re-evaluation of human expertise in the AI era, which are critical areas for pedagogical research

While this study focuses on pedagogical design, the broader impact of AI on higher education may suggest a shift from reactive adaptation to proactive integration. As such AI can be viewed not only as a disruptive innovation to be managed, but also as a tool to be intentionally integrated with forward-thinking designs that anticipate evolving AI affordances (Carvalho et al., 2022; O'Dea, 2024). This transition may extend beyond pedagogy to require significant curriculum redesigns that address the evolving demands of an AI-driven society and labor market (Abbasi et al., 2025; Jaramillo & Chiappe, 2024; Liang et al., 2025). In addition, this transition may have institutional implications, challenging the longevity of fixed syllabi in favor of adaptive as AI has the potential to open different learning pathways (Jose et al., 2025). On the other hand, this transition presents risks, including the potential for epistemic decline and cognitive atrophy as learners outsource critical judgment (Kosmyna et al., 2025; H.-P. (Hank) Lee et al., 2025), the propagation of inherent AI biases (Bender et al., 2021), unresolved questions of academic authenticity (Nguyen, 2025; Wang et al., 2024; Yusuf et al., 2024), and other risks reviewed by Sengar et al. (2024). Therefore, instructors and institutions may consider adopting a dual strategy by implementing immediate adaptations such as revised assessments and AI literacy, while proactively preparing for a future defined by dynamic human-AI collaboration. While this future is speculative, multimodal AI that was emerging (Sengar et al., 2024) is now an active reality. This highlights the importance of anticipating how AI capabilities will evolve at different intervals and proactively planning pedagogical strategies and curriculum development in alignment with those trajectories (Jaramillo & Chiappe, 2024; Walter, 2024). Thus, instead of chasing a moving target, educational design may benefit from examining potential AI development trajectories to empower learners with transferable competencies that will endure in an AI-driven future.

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